Loop Naming Rules

A loop name contains exactly 7 characters. The underscores are considered characters and are required. A loop name is found by selecting one item from each column in Table #1 (Roadway, Direction, Lane type, Lane #). These items are then concatenated into a loop name. The correct item is found by following the conventions in the list of rules below. Special loops that need to be identified to the 170 are shown in Table #2. Speed loops must also be identified to the 170. These loops are identified by an uppercase 'S' in column 6. The corresponding loop for the trap will match all columns except 6, which will contain an '_' and column 1, which will not contain an 'M'. Note: DeviceTypes 'T' & 'R' are reserved for system use and are not used when creating a loop name.

Table #1

RoadwayClass	RoadwayType	Direction	Lane Class	Lane Type	Device type	Lane #
_-GP	M-mainline	S -southbound	_-shared, thru	_-mainline	_-not S, T, R	1
A-Auxiliary	C-CD	N -northbound	R-Right Shoulder/(Adv. Q)	A-Adv. Q	S-Speed loop	2
M-Metering	R-reversible	E -eastbound	L-Left Shoulder/(Adv. Q)	D-demand	T-speed Trap	3
D-Duplicate	A-Arterial	W -westbound	H-HOV / HOT	I-inter Q	R-meter Rate	4
O-Off system	X-Cross St.		B-Bus Only	O-On	C-Coupled Speed	5
T-toll/tag reader			Y-Bicycle Only	P-passage		6
0-9-Border Wait			V-metered HOV	Q-queue		7
			W-metered HOV on shoulder	X-exit		8
			Z-HOV on shoulder	M-merge		9

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Table #2.

Passage	Demand
Queue	Intermediate Queue
Adv. Q Left	Adv. Q Right
HOV demand	HOV Passage
Meter station loops	

Roadway Naming Rules

1) Each cabinet is assigned a principal roadway. If a cabinet on I-5 (principal roadway) has loops on I-90, then the I-90 loops are considered auxiliary and an "A" is added to the roadway name (col. 1). In the case of on-ramps and exits, the principal name will prevail. If a loop could be called an exit from I-90 and an on-ramp to I-5, then it would be called an on-ramp.
2) Loops that are to be part of a ramp metering station will have a roadway type beginning with an "M". Speed loops and HOV lanes typically DO NOT have the roadway type beginning with an "M".
3) All loops must have a roadway type. On-ramps and exits will assume the roadway type of the roadway that the ramp enters or leaves.
4) At the border crossings, there may be several stations of loops connected to a single cabinet for queue detection. The leading number in the loop name indicates which station the loop resides in.

Direction Rules

1) All loops will have a direction code: N - northbound, S - southbound, E - eastbound, W - westbound.
2) Reversible roadways will use the direction of increasing milepost. The I-5 express lanes are considered as Northbound.

Lane Naming Rules

1) All loops will have one lane type.
2) Each metered lane can have up to two advanced queue loops, one left movement and one right: "LA" and "RA". If no movement is associated with the advanced queue loop, then use "RA".
3) Depending on the ramp length, each metered lane can have up to 2 queue loops: one intermediate queue loop "_I" and one queue loop "_Q". The queue loop and the intermediate queue loop will be evenly spaced, 200 to 500 feet apart, depending on the ramp length. The intermediate queue loop will be placed between the queue loop and the advanced queue loops.
4) Use "_X" for exit ramps and "_O" for on-ramps that are not metered.
5) HOV loops are specified by an "H" as the first lane type letter ("Z" if the HOV lane is on the shoulder).
6) Use "__", "H_" or "Z_" for mainline loops.
7) Use "V" or "W" if the HOV lane is metered.

Numbering Rules

1) All loops of similar type are numbered from upstream to downstream.
2) All multi-lane roadways will be numbered from right to left while looking downstream (in the direction of traffic flow). If one or more lanes are HOV, they will be numbered as if they are general purpose lanes (add 1 to the adjacent lane number).
3) Each HOV lane bypass will have the same lane number as the ramp lane that it bypasses. (EXCLUDING versions 5.70 & 5.170 and later)
4) Each metered HOV lane will have the next lane number (sequential) from the adjacent metered ramp lane number. (version 5.70 & 5.170 and later)
5) Speed loops will have an "S" before the lane number (column 6).
6) Ramp meter lanes will be numbered from right to left and upstream to downstream. All existing (installed before 1992) ramp meter lanes are numbered "2".

Station Naming Rules

1) All station names will contain a roadway and direction type followed by either "_Stn", “MStn”, “RStn”, “XStn”, or "HStn".
2) The naming scheme is for mainline, CD, reversible, auxiliary and exit or HOV lanes on these roadway types. Choose one from each column in Table 3.
3) The default station equation generator looks at the 1st, 2nd & 3rd characters in the loop name. D* loops are ignored.
4) User created stations will have a “U”, “W” or “D” in the first column.
   Note: “D” will only be used as a last resort (i.e. multiple U* stations on a given roadway type and direction).

Table #3

Designator	Roadway	Direction	Lane Type
_ -mainline	M -mainline	S –southbound	_ -mainline
A –auxiliary	C -CD	N –northbound	X -exit ramp
M –metered	R –reversible	E –eastbound	H -HOV
D –duplicate	Y –ATM case	W –westbound	M –ATM mainline
U –user defined			R –ATM ramp
W –user defined.			

Special Cases

Roundabout Loops